12 kW Wind Turbine for Iceland: Magnet Variant N42SH 30 × 15 × 6 (P1 generator)

Status: **Rev A, proposed** · 2026-11-05 (at Martin's request: "adjust the design to 30x15x6 N42SH and compare it to the original design")
Owner: Category 4 (Generator) with 10 (Manufacturing and quality)
Baseline: 19 – Final magnet geometry Rev B (40 × 20 × 7.5, N40EH / N40UH, approved 2026-11-05). Doc 19 stays the approved drawing set; this document is an alternative rotor that can be built instead of it.
Analysis: `analysis/generator/sh30.py` `S1d` → `sh30_S1d.json`; `magnets.py` `S1d` (eddy loss, ripple, forces) → `magnets_summary.json`
Geometry: `cad/generator_mfg/magnet_geom_s30.py` (every dimension below comes from here)
Pictures: `cad/generator_mfg/figures_s30.py` → D01 to D04, DI01

Summary

- **Layout:** each pole is one stack of two N42SH 30 × 15 × 6 blocks bonded N face to S face (a 30 × 15 × 12 magnet), five stacks along the 74 mm stator, the rows step-skewed 0.20° apart. 10 blocks per pole, **400 per generator** (+ 20 spares), 8.21 kg of magnet.
- **It matches Rev B** once the coils get **29 turns instead of 28**: 378 V at 100 °C (Rev B 380 V), 784 Nm at the same phase current (Rev B 787 Nm), copper loss **+8 %** at rated torque, cogging 0.02 Nm, and a much cleaner voltage (phase THD 2 % against 13 %).
- **Demagnetisation:** the 12 mm magnet keeps **32 % margin at 100 °C** and 13 % at 120 °C in the transient short circuit. That is clearly better than N42SH on the Rev B drawing (20 % and −1 %), and keeps the 20 % rule up to about **112 °C**. It still does not meet the original 120 °C case, so it needs the 100 °C design basis and the three conditions of doc 19 §16.3.
- **Why build it:** 30 × 15 × 6 N42SH is a stock catalogue part at European and US makers, the grade needs little Dy/Tb, and the small pieces have a third of Rev B's eddy loss (4.7 W against 15.5 W).
- **What it costs:** 2.5 × the pieces and a pair-bonding step, a new drum seat Ø628.05 (drum steel under the magnets 9.0 mm instead of 13.7 mm, running at about 1.7 T), new comb bars and 29-turn coils.

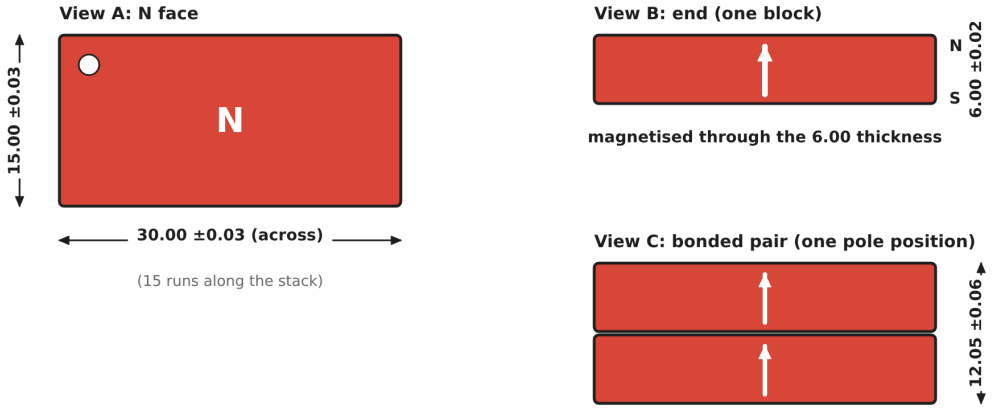
Recommendation: keep Rev B as the production baseline wherever N40EH or N40UH can be bought. Use this variant when only SH is available outside Asia; it is the better of the two SH options (more margin than N42SH 40 × 20 × 7.5, at the price of more parts and a new seat and winding).

1. Why this layout

Doc 19 §15 looked at single 6 mm blocks. One block across × five along (S1) builds easily but gives only 312 V; three across × three along (S3) gives 376 V but leaves 2 mm between poles and overhangs the stack 8 mm at each end. The way out is to **stack two blocks radially**: the S1 layout with a 12 mm magnet. The longer magnet gives the flux back (narrower pole, but 12 mm instead of 7.5 mm magnet length against the same 2.5 mm magnetic gap) and raises the operating point, which is what gives the better demagnetisation margin. The 0.63 pole span cogs at 5.4 Nm unskewed; five rows 0.20° apart (one fifth of the 1° cogging period) cancel it.

D01 Detail drawing NP1-220S: N42SH block 30 × 15 × 6 and the bonded pair

Drawn 10× real size. Dimensions in mm. Two blocks are bonded N face to S face into one 12 mm pair (shop step).



N face of one block on the S face of the other:
they pull together. Thin epoxy 0.02 to 0.05, edges flush ±0.05.

Flatness of N and S faces 0.02
Parallelism N to S face 0.02
Chamfer 0.2 to 0.3 × 45°, all 12 edges
Magnetic axis within 2° of the face normal

Part	NP1-220S N42SH block, catalogue 30 × 15 × 6
Material	Sintered NdFeB N42SH: Br ≥ 1.29 T, HcJ ≥ 1,592 kA/m (20 kOe)
Coating	NiCuNi ≥ 15 µm + epoxy ≥ 15 µm (in the sizes)
Mark	White dot Ø3 on the N face
Mass	20.5 g each
Quantity	400 + 20 spare per generator (200 pairs)
Moment	Each piece within ±1.5 % of the lot mean, value per serial

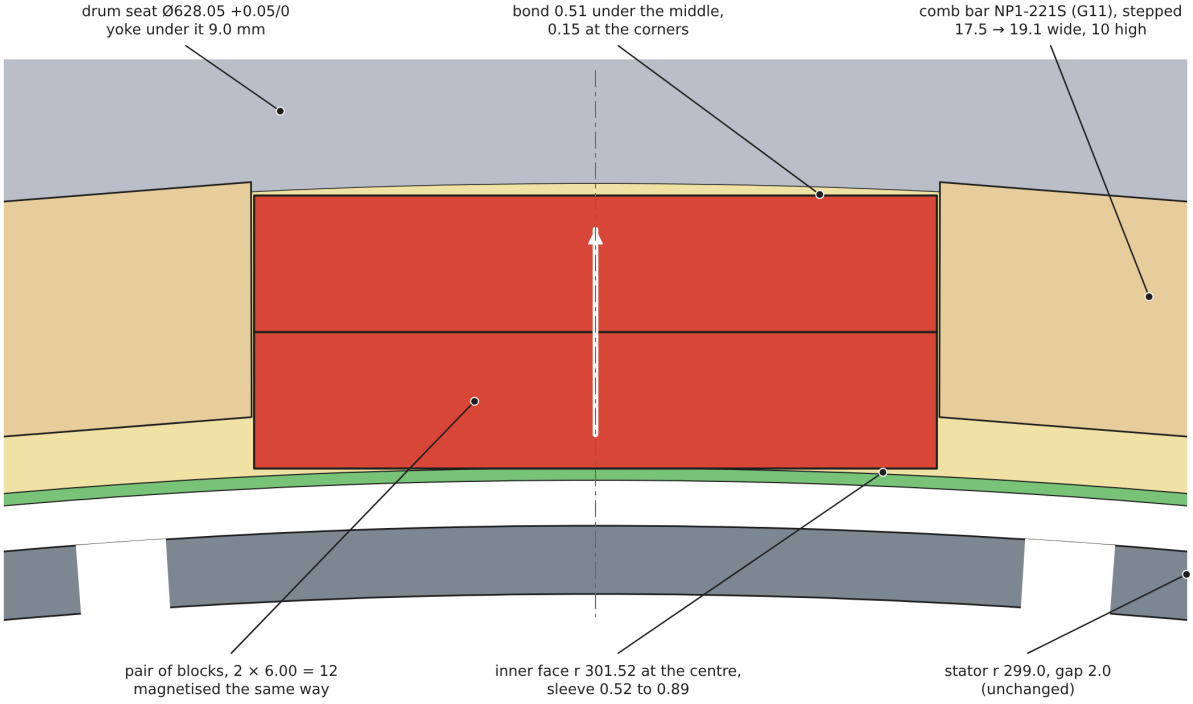
Doc 20 · Wind turbine 12 kW · P1 generator, N42SH variant

2. Geometry

Item	Value
Block	N42SH 30 × 15 × 6 mm, magnetised through 6, NiCuNi (as doc 19 §8)
Magnet per pole	1 across × 5 along × 2 high = 10 blocks; pairs bonded N to S
Rows	5 × 15 + 4 × 0.20 shim = 75.8 mm, centred on the stack (0.9 overhang each end)
Row positions (front face, x)	342.1 / 357.3 / 372.5 / 387.7 / 402.9
Step skew	−0.4 / −0.2 / 0 / +0.2 / +0.4° front to rear (0.80° in all)
Drum seat	**Ø628.05**, land x 335 to 425 (doc 18 land length unchanged)
Bond line	0.15 at the block edges, 0.51 under the block middle
Magnet inner face	r 301.52 (sleeve bore r 301.0: 0.52 in the middle, 0.89 at the edges)
Pole span	5.70° (0.63 of the 9° pitch)
End rings	7.1 wide at each end
Comb bars (NP1-221S)	G11, one stepped bar per pole gap, 82 long, 10 high, 17.54 wide at the top, 19.11 at the seat
Drum steel under the seat	9.0 mm (Rev B 13.7 mm)
Mass	20.5 g per block, 8.21 kg per generator
Pieces	400 + 20 spares

D02 One pole of the N42SH variant (cross-section, 20× real size)

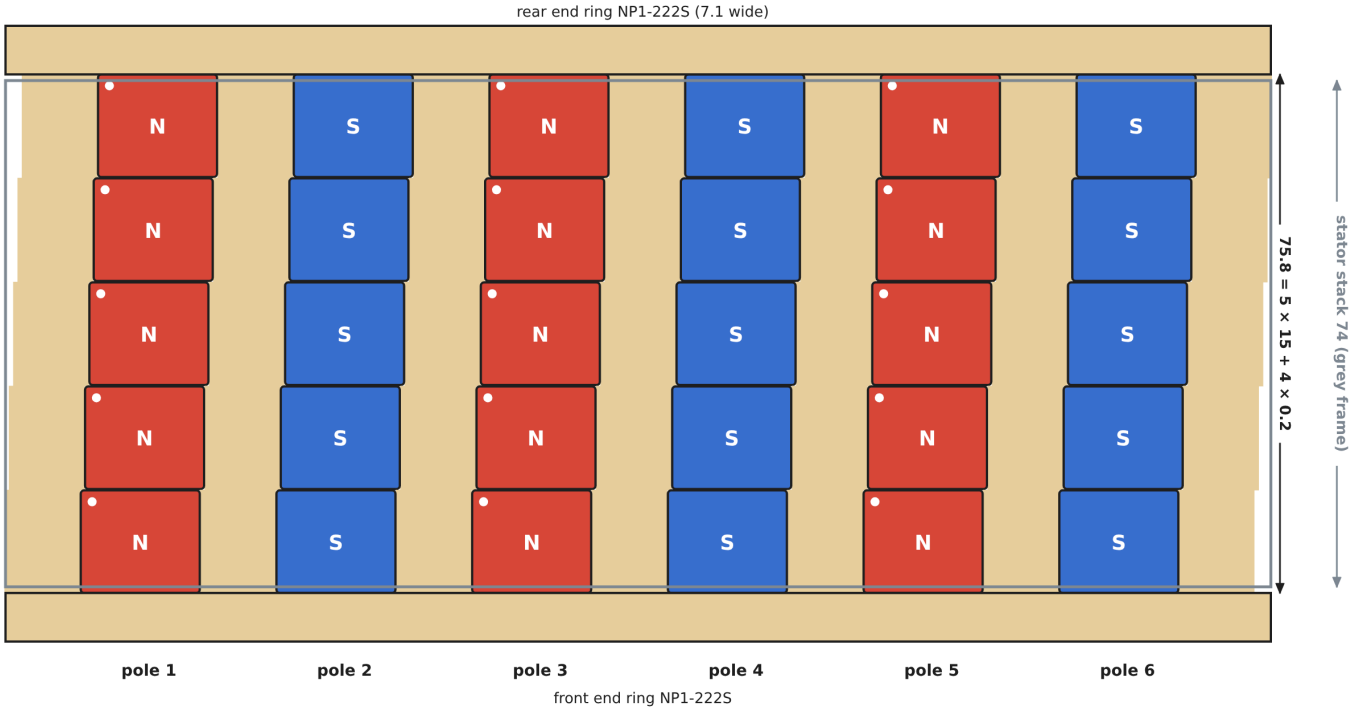
One 30 mm wide pair of blocks (12 mm) per pole, wide comb bars between the poles.



Pole pitch 9.000°, pole span 5.70° (0.63), block 30 wide
Five rows along the stack, turned 0.20° from row to row
(front −0.4° … rear +0.4° from the drawn centre)

D03 Where every block goes (N42SH variant): drum seat unrolled

6 of the 40 poles. Each pole = 5 rows along the stack, each row a pair of blocks. Rows step 0.20° (1.1 mm).

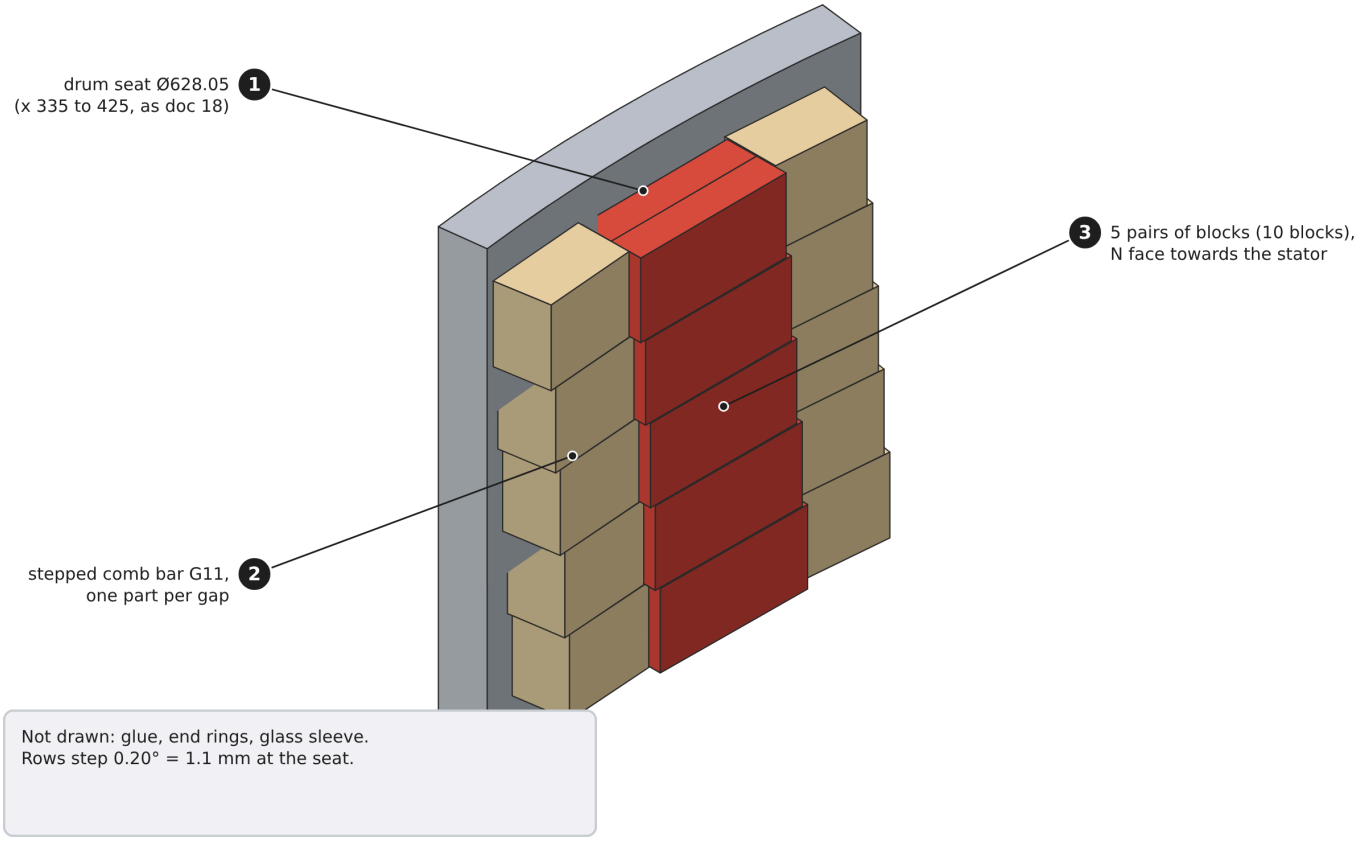


tan = G11 stepped comb bars and end rings

- rows turned 0.20° from one to the next (0.80° front to rear)
- the comb bar is ONE stepped G11 part per gap: the steps set the skew
- white dot visible = N pole; N and S alternate

DI01 One pole of the N42SH variant (isometric), seen from the stator side

Five rows of bonded pairs on the drum seat, each row turned 0.20° further. Stepped comb bars either side.



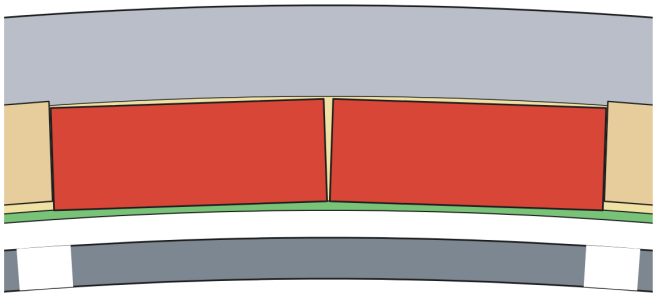
Not drawn: glue, end rings, glass sleeve.
Rows step 0.20° = 1.1 mm at the seat.

3. Comparison with Rev B

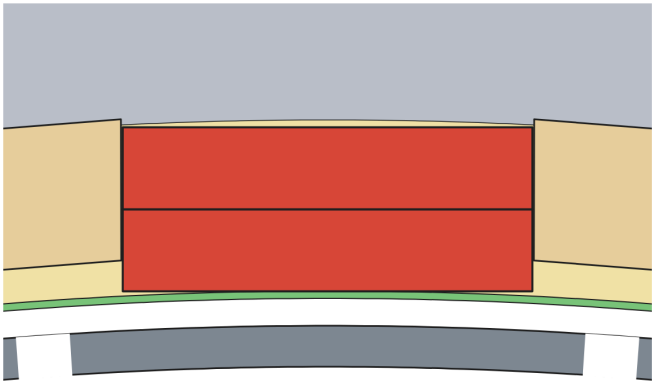
D04 Rev B (40 × 20 × 7.5) and the N42SH variant (30 × 15 × 6 pairs)

Same stator, gap, sleeve and drum outside. One pole cut across, 12× real size.

Rev B: 2 blocks 20 wide, 7.5



N42SH variant: 1 pair 30 wide, 12



	Rev B (doc 19)	N42SH variant (doc 20)
Block	40 × 20 × 7.5	30 × 15 × 6, bonded in pairs
Grade	N40EH / N40UH	N42SH
Pieces per generator	160 + 8	400 + 20
Magnet mass	7.30 kg	8.21 kg
Pole span / rows / skew	0.85 / 2 / 0.50°	0.63 / 5 / 4 × 0.20°
Drum seat / yoke under it	Ø618.65 × 96 / 13.7 mm	Ø628.05 × 90 / 9.0 mm
Turns per coil	28	29
Line EMF, 100 °C	380 V	378 V
Torque at the same phase current	787 Nm	784 Nm
Copper loss at rated torque	reference	+ 8 %
Cogging (skewed)	0.01 Nm	0.02 Nm
Magnet eddy loss, rated	15.5 W	4.7 W
Worst short circuit margin, 100 / 120 °C	N40EH –/47 %, N40UH –/30 %	32 / 13 %
Sourcing	Dy/Tb heavy, mostly China	SH: little Dy/Tb, EU/US makers

	Rev B (doc 19)	N42SH variant
Block / grade	40 × 20 × 7.5, N40EH or N40UH	30 × 15 × 6 in pairs, N42SH
Pieces / magnet mass	160 + 8 / 7.30 kg	400 + 20 / 8.21 kg
Pole span / rows / skew	0.85 / 2 / 0.50°	0.63 / 5 / 4 × 0.20°
Drum seat / steel under it	Ø618.65 × 96 / 13.7 mm	Ø628.05 × 90 / 9.0 mm
Turns per coil	28	**29**
Line EMF at 178 rpm, 100 °C	380 V	378 V
Torque at the same phase current	787 Nm	784 Nm
Copper loss at rated torque	reference	+8 %
Cogging, skewed (unskewed)	0.02 (6.6) Nm	0.02 (5.4) Nm
Torque ripple at rated load, skewed	2.0 %	2.3 %
Phase / line THD, no load	13.4 / 0.9 %	2.0 / 1.8 %
Magnet eddy loss, rated	15.5 W	4.7 W
Radial magnet pull per pole, no load / rated	37 / 131 N	50 / 145 N

	Rev B (doc 19)	N42SH variant
Transient SC margin, 100 / 120 °C	N40EH > 47 / 47 %; N40UH > 30 / 30 %	**32 / 13 %**
20 % margin kept up to	> 150 °C (EH), ≈ 130 °C (UH)	≈ 112 °C
Grade availability outside Asia	EH/UH: Dy/Tb heavy, few quotes	SH: stock grade at Neo, VAC and others

For reference, N42SH in the Rev B block (doc 19 §16) gives 388 V with 28 turns but only 20 % / −1 % margin at 100 / 120 °C. The variant buys 12 points of margin with the thicker magnet.

What changes in practice:

- **Winding: 29 turns.** With 28 turns the variant gives 365 V and 757 Nm (−4 %). One more turn per coil fits the slot (the copper area is unchanged, wire one step thinner or the same wire at a slightly higher fill) and brings both back. Same ampere-turns at the same torque, so the slot heat goes up by the 8 % from the 3 % lower torque per ampere-turn and the longer wire: hot spot about 2 to 3 K higher than doc 06.
- **Voltage.** The worst-case peak (20 °C magnets, overspeed) is about 890 V, slightly under Rev B, still inside the 1,200 V parts.
- **Drum steel.** The seat grows by 9.4 mm in diameter, so the drum shell under the magnets is 9.0 mm thick. The FEA includes the S355 drum: at no load it runs at 1.45 T median and 1.7 T at the pole edges (Rev B 0.9 and 1.4 T). The EMF above already includes this saturation. Mechanically the drum must be rechecked for the 11 % higher pull at rated load on a thinner shell (doc 18 drum check, open item O3).
- **Rotor field and eddy loss.** Five 15 mm pieces along the stack cut the eddy loss to 4.7 W; the magnets run cooler than Rev B for the same load.

4. Demagnetisation

Worst element against N42SH catalogue HcJ (1,592 kA/m at 20 °C, −0.60 %/K), 29 turns make no difference here because the fault current scales with them:

Magnet temperature	HcJ	Rated load	Steady short circuit	Transient short circuit (2 ×)
60 °C	1,210 kA/m	69 %	61 %	51 %
80 °C	1,019 kA/m	64 %	55 %	43 %
100 °C	828 kA/m	56 %	47 %	**32 %**
120 °C	637 kA/m	44 %	32 %	13 %

Everything except the transient short circuit passes at 120 °C. The variant therefore takes the doc 19 §16.3 design basis: **100 °C design case**, released only if the bench test measures ≤ 75 °C magnets in the worst case, infrared drum watch with derating at 85 °C and a pitch stop at 95 °C, and no deliberate phase shorting. With those, it has 12 K more reserve than N42SH 40 × 20 × 7.5.

5. Buying

- **Part:** 30 × 15 × 6 mm, N42SH, magnetised through 6, NiCuNi 10 to 20 µm, tolerances ±0.05 on 30 and 15, ±0.03 on 6 (the pair stack height matters more than a single block), Br and HcJ certificates per batch, as doc 19 §8 otherwise. 420 per generator.
- **Sources:** this is a catalogue size at most makers; the outside-Asia list of doc 19 §16.4 (Neo Narva, VAC Hanau / eVAC, Noveon) applies. Small lots go through distributors at 2 to 3 × the Chinese price.
- **Ask for the pairs pre-bonded** if the maker offers it (many bond stacks before magnetising, which is easier and safer than bonding two magnetised blocks). Otherwise the pair is bonded in-house (§6).

6. Changes to build and documents

What	Change
Drum (doc 18 §6.4)	Seat Ø628.05 on the same land x 335 to 425; drum stiffness recheck with 9.0 mm shell
Comb bars NP1-221S	New stepped G11 bar, 82 × 10 high × 17.54 / 19.11, one per pole gap (40)
End rings	7.1 wide
Pair bonding (new step before doc 19 §9)	In a non-magnetic V-jig: the second block is pushed on from the side (never let them snap together), 0.02 to 0.05 glue line, pair height checked 12.00 ±0.08
Placing jig	Five row stops 15.2 apart, offset 0.20° per row (template as doc 19 §9, new insert)
Stator coils (doc 18 §5)	29 turns per coil
Sleeve check (doc 19 Q1)	Gap to the sleeve bore up to 0.9 at the block edges (range extended from 0.8)
Controller	Magnet temperature watch, derate 85 °C, stop 95 °C (doc 19 §16.3)

7. Open items

#	Item	Owner
O1	Bench test: magnet temperature ≤ 75 °C at rated and 110 % load and after hot shutdown	Cat 4
O2	Coil with 29 turns: check slot fill with the chosen wire, update doc 06 thermal model (+2 to 3 K expected)	Cat 4
O3	Drum with 9.0 mm shell under the magnets: stiffness and stress under magnet pull (145 N radial per pole at rated load)	Cat 5
O4	Quote pre-bonded pairs from Neo / VAC; otherwise prove the in-house pair bonding on 20 pairs	Cat 10
O5	3D check of the row-end flux (five short rows, 0.9 overhang); the 2D model ignores end effects as in doc 19	Cat 4

Appendix A. How to rerun

```
cd analysis/generator
python3 sh30.py S1d          # EMF, cogging, torque, demag   -> results/sh30_S1d.json  (~15
min)
python3 magnets.py S1d       # eddy loss, ripple, forces     -> results/magnets_summary.json
cd ../../cad/generator_mfg
python3 magnet_geom_s30.py   # geometry table
python3 figures_s30.py --png # D01 to D04, DI01
```